A Metacognition-Based Empathetic Smart Home

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***Abstract*—A household-assistance agent must judge whether an action helps the resident at that moment as well as perform the task. Whether a cup should be cleared depends on reuse intentions, while asking the resident about every missing detail creates interaction burden. This project proposes a smart-home agent combining metacognition, which distinguishes established facts from missing information, with empathetic intelligence, which considers resident intentions, living context, and intervention burden. The agent selects necessary observations and questions, compares the predicted outcomes of acting and waiting, and reconsiders decisions when execution outcomes differ from predictions. Team members capture requested scenes, and selected actions are executed in a digital twin. The representative task checks end of use and authorization, empties remaining drink when needed, loads and starts a dishwasher, and verifies washing of the target cup. While waiting for the dishwasher, the agent retains the original goal, checks other dishes, and prepares the next cycle. The scope also covers cooktop safety, return-home reporting, and planning for novel combinations. Comparative evaluation examines task completion, incorrect executions, and the burden of questions and capture requests.**

Role Assignments (till the middle of the term)

The team members' roles and primary responsibilities are as follows.

Roles	Name	Task description and etc.
User	Jihun Shin	Provide household scenarios, intentions, exceptions, and usability feedback.
Customer	Junhyeong Kim	Review the problem, scope, priorities, and acceptance criteria.
Software developer	Seokjun Park	Specify inputs, outputs, requirements, constraints, and verification cases; subsequently design, implement, and test.
Development manager	Jeong-O Heo	Coordinate schedules, progress, changes, integration, and deliverables.

I. INTRODUCTION

Household assistance requires considering both the state of objects or appliances and the resident's intentions. A resident who leaves a cup may intend to drink again, and one who leaves the kitchen may still be cooking. Clearing a visible cup or switching off a cooktop solely because nobody is nearby can disrupt ongoing activities. Conversely, waiting only for commands can miss opportunities to help, while referring every decision to the resident requires continued instruction and supervision. These situations call for judgment about when help is needed, what further information to obtain, and when to act or wait.

A. Motivation and Key Concepts

In this project, metacognition is the ability to distinguish established facts from unknown information, select necessary observations and questions, and revise judgments when execution outcomes differ from predictions. When a cup's interior is hidden, the agent requests another viewing angle. It asks the resident about reuse intentions that an image cannot establish and carries valid previous answers into subsequent decisions.

Empathetic intelligence is the ability to choose how and when to help by considering resident intentions, living context, and intervention burden. The agent leaves a cup intended for reuse in place and considers disruption when requesting nonurgent clarification. It judges the appropriateness of acting or asking now alongside task feasibility. Metacognition supports necessary checks and revision of judgments, while empathetic intelligence supplies criteria for fitting these to the resident's life. The project adopts metacognition-based empathetic intelligence to connect these abilities.

The objective is a prototype of the decision-making component of a prospective CLOiD system that respects residents' activities and carries authorized work through to completion. It repeats observation, clarification, prediction, selection, execution, and outcome checking, using accumulated experience to improve checks and planning. Team members capture requested scenes, and a digital twin presents the selected actions and outcomes.

B. User Needs and Deliverables

The intended resident-facing outcomes are:

- Action consistent with intentions and authorization, or waiting with a stated reason.
- Interaction that reuses valid information and requests only necessary clarification.
- A report of whether the target cup has been washed, remains unfinished, or is waiting, with supporting reasons.

Planned deliverables are a prototype demonstrating this process, records linking observations to outcomes, comparative evaluation and failure cases, and reproduction instructions.

C. Related Work

KnowNo calibrates planning uncertainty and requests human help when needed [1]. PACT uses observations and interaction history to decide whether clarification is needed and evaluates accuracy together with clarification burden [2]. This course project connects these concepts through observation requests, authorization checks, and virtual planning and

execution for the defined household tasks, evaluating task outcomes alongside intervention burden.

II. PROJECT SCOPE

A. System Boundary

The development target is the decision-making component of a prospective CLOiD system. Real-scene demonstrations use requested images or video and user answers, with capture supported by a team member. Actions occur in the virtual home; physical robots and appliances are outside the execution scope.

B. Scenarios

1) Cup cleaning: authorized disposal, dishwasher loading and activation, and completion verification for the relevant cycle.
2) Cooktop safety: notification, waiting, or virtual shutdown based on context, return intention, responses, and prior authorization.
3) Return home: summarizing virtual actions and deferred work after the resident returns.
4) Novel combinations: selecting observations, questions, actions, and waiting under new combinations of cups, intentions, and appliance states.

Experience will be collected in a virtual living environment with simulated residents and an observation interface, followed by a before-and-after learning comparison. Hidden truth is not directly supplied to online decisions.

Excluded topics are unrestricted household work, invention of new action types, conflicting intentions among multiple residents, and laundry. Long-duration plans across diverse chores and complex changes to actions already in execution are deferred.

III. REQUIREMENTS

A. Common Behavior

The system shall identify assistance opportunities and missing information, request necessary observations or answers, compare alternatives using available information and authorization, and select virtual execution or waiting. It shall record checks, decision reasons, outcomes, and prediction discrepancies, explain decisions when asked, and distinguish virtual outcomes from physical-scene changes.

B. UC-01: Continuous Cup-Cleaning Flow

The use case begins when a resident leaves a cup and moves away. This does not establish that use has ended. After checking contents, reuse intention, and authorization, the agent empties remaining drink when appropriate, loads the dishwasher, and starts it. The goal is complete only after confirming washing of that cup.

If a cycle is running, the agent retains the original goal while searching for other dishes and checking observations, intentions, and authorization. After that cycle finishes, it rechecks space and availability and washes prepared original and additional authorized items in a subsequent cycle. Additional search shall not indefinitely defer the original goal.

C. Cup-Cleaning Functional Requirements

CUP-01. The system shall interpret the target cup and its usage context from input observations and distinguish information that cannot be determined.

CUP-02. The system shall generate an observation request identifying the information needed and the purpose of the request when cup-cleaning information is insufficient.

CUP-03. The system shall ask the resident when intention or authorization is required and incorporate the answer into its decision.

CUP-04. The system shall compare alternatives, including cleaning and waiting, using the information obtained and applicable authorization.

CUP-05. The system shall not virtually clear a cup that is in use, intended for reuse, or lacks the required cleaning authorization.

CUP-06. The system shall reconsider an unexecuted plan when relevant new information arrives before execution.

CUP-07. The system shall execute the selected virtual plan, including emptying remaining drink when needed, loading the dishwasher, starting a cycle, and washing, while distinguishing progress, success, and failure.

CUP-08. The system shall record observation requests, questions, answers, decisions, reasons, and results, and explain a decision from that history when asked.

CUP-09. The system shall report completion only after confirming the completion of the virtual wash cycle that contains the target cup.

CUP-10. The system shall confirm that use has ended and that required disposal is authorized before emptying remaining drink and loading the cup.

CUP-11. The system shall check the necessary availability conditions before loading or starting the dishwasher.

CUP-12. The system shall retain the original cup's unfinished goal, confirmed information, authorization, and waiting reason while the dishwasher is busy.

CUP-13. The system shall be able to search for other dishes while waiting and establish subsequent-cycle candidates through necessary observations, usage-intention checks, and authorization.

CUP-14. The system shall recheck availability after the current cycle finishes and resume washing prepared items in a subsequent cycle.

CUP-15. The system shall recheck each item's current conditions before the subsequent cycle and shall not repeatedly defer a prepared original goal solely because additional search or candidate answers remain unfinished.

IV. VERIFICATION PLAN

A. Acceptance Cases and Comparisons

Cup-cleaning verification covers 18 acceptance cases addressing normal washing, reuse, missing authorization, unavailable observations, changes before execution, nonresponse,

failure, incorrect attribution of an earlier cycle, and search and resumption while waiting.

The evaluation will examine whether necessary checks reduce incorrect executions relative to acting without clarification, and whether selective requests reduce burden relative to asking about all information without increasing errors. A comparison omitting prior intentions and learning is also planned.

Comparisons start from the same initial observations, public information, and allowed actions. Records include incorrect executions and authorization violations, questions, capture requests, repeated requests for the same information, and completed, waiting, or failed outcomes. Cup completion is determined by linking the target to its actual virtual wash cycle. Planning success on new combinations and explanation comprehension will also be examined.

Evidence will link observations and answers, established facts and unknowns, request purposes, alternatives and decision reasons, execution results, and completion judgments. Learning will be compared before and after experience using situations excluded from training, with evaluation designed to distinguish reuse of intention records from learned changes.

B. Validation Scope

Real-scene trials will assess media interpretation and request-response interaction, while the virtual living environment will assess planning, goal retention, washing completion, and learning. Physical robot and appliance control performance is outside the scope of this evaluation.

REFERENCES

[1] A. Z. Ren et al., *Robots That Ask For Help: Uncertainty Alignment for Large Language Model Planners*, arXiv:2307.01928, 2023. [Online]. Available: https://arxiv.org/abs/2307.01928

[2] C. He et al., *PACT: Proactive Asking for Continual Task Assistance in Human-Robot Collaboration*, arXiv:2605.24350, 2026. [Online]. Available: https://arxiv.org/abs/2605.24350